Code Quality & Security Audit Report

Comprehensive Static Code & Dependency Analysis

PROJECT	DART SDK	TARGET PLATFORMS	STATE / NETWORK
flutter_base	^3.4.4	android, ios	setState / dio

1. Executive Summary & Category Breakdown

CATEGORY	HIGH SEVERITY	MEDIUM SEVERITY	LOW SEVERITY	TOTAL ISSUES
Security & Secrets	33	0	1	34
Performance & UI Lag	23	0	0	23
Memory & Lifecycle Leaks	37	0	0	37
Maintainability & Class Size	1	44	0	45
Total Detected Issues	**94**	**44**	**1**	**139**

2. Critical Security Findings

RULE ID	SEVERITY	FILE LOCATION	LINE	FINDING DETAILS & RISK
`SEC-SEC-001`	CRITICAL	`ios/Runner/AppDelegate.swift`	12	**Hardcoded API Key**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` *Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.*
`SEC-SEC-001`	CRITICAL	`ios/Runner/GoogleService-Info_old.plist`	6	**Hardcoded API Key**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ```

RULE ID	SEVERITY	FILE LOCATION	LINE	FINDING DETAILS & RISK
				Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.
`SEC-SEC-001`	CRITICAL	`android/app/google-services_old.json`	18	**Hardcoded API Key**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` *Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.*
`SEC-SEC-001`	CRITICAL	`android/app/src/main/AndroidManifest.xml`	81	**Hardcoded API Key**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` *Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.*

RULE ID	SEVERITY	FILE LOCATION	LINE	FINDING DETAILS & RISK
`SEC-SEC-001`	CRITICAL	`lib/utils/common/constants.dart`	169	**Hardcoded API Key**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` *Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.*
`SEC-SEC-001`	CRITICAL	`lib/utils/common/constants.dart`	191	**Hardcoded API Key**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` *Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.*
`SEC-SEC-001`	CRITICAL	`lib/utils/common/constants.dart`	191	**Hardcoded API Key**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` *Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.*
`SEC-SEC-001`	CRITICAL	`lib/utils/common/constants.dart`	198	**Hardcoded API Key**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-

RULE ID	SEVERITY	FILE LOCATION	LINE	FINDING DETAILS & RISK
				engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` *Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.*
`SEC-SEC-001`	CRITICAL	`lib/utils/common/constants.dart`	199	**Hardcoded API Key**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` *Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.*
`SEC-SEC-001`	CRITICAL	`lib/utils/common/constants.dart`	203	**Hardcoded API Key**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` *Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.*
`SEC-SEC-001`	CRITICAL	`lib/utils/common/constants.dart`	207	**Hardcoded API Key**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ```

RULE ID	SEVERITY	FILE LOCATION	LINE	FINDING DETAILS & RISK
				Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.
`SEC-SEC-001`	CRITICAL	`lib/utils/common/constants.dart`	221	**Hardcoded API Key**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` *Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.*
`SEC-SEC-001`	CRITICAL	`lib/utils/common/constants.dart`	221	**Hardcoded API Key**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` *Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.*

RULE ID	SEVERITY	FILE LOCATION	LINE	FINDING DETAILS & RISK
`SEC-SEC-001`	CRITICAL	`lib/utils/common/constants.dart`	222	**Hardcoded API Key**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` *Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.*
`SEC-SEC-001`	CRITICAL	`lib/utils/common/constants.dart`	222	**Hardcoded API Key**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` *Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.*
`SEC-ATH-001`	HIGH	`lib/utils/common/constants.dart`	164	**Potential Hardcoded Password or Auth Key**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` *Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.*
`SEC-ATH-001`	HIGH	`lib/utils/common/constants.dart`	165	**Potential Hardcoded Password or Auth Key**: Hardcoded credentials are static values that get compiled directly into the application

RULE ID	SEVERITY	FILE LOCATION	LINE	FINDING DETAILS & RISK
				package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` *Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.*
SEC-ATH-001	HIGH	lib/utils/common/constants.dart	169	**Potential Hardcoded Password or Auth Key**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` *Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.*
SEC-ATH-001	HIGH	lib/utils/common/constants.dart	191	**Potential Hardcoded Password or Auth Key**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` *Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.*
SEC-ATH-001	HIGH	lib/utils/common/constants.dart	198	**Potential Hardcoded Password or Auth Key**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they

RULE ID	SEVERITY	FILE LOCATION	LINE	FINDING DETAILS & RISK
				can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` *Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.*
`SEC-ATH-001`	HIGH	`lib/utils/common/constants.dart`	199	**Potential Hardcoded Password or Auth Key**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` *Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.*
`SEC-ATH-001`	HIGH	`lib/utils/common/constants.dart`	520	**Potential Hardcoded Password or Auth Key**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` *Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.*
`SEC-ATH-001`	HIGH	`lib/utils/helpers/shared_pref_helper.dart`	12	**Potential Hardcoded Password or Auth Key**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const

RULE ID	SEVERITY	FILE LOCATION	LINE	FINDING DETAILS & RISK
				String stripeApiKey = "sk_live_51H..."; ``` *Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.*
`SEC-ATH-001`	HIGH	`lib/utils/helpers/shared_pref_helper.dart`	22	**Potential Hardcoded Password or Auth Key**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` *Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.*
`SEC-WV-001`	HIGH	`lib/reusable_component/cms_page.dart`	1	**Unsafe WebView Configuration**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` *Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.*
`SEC-WV-002`	HIGH	`lib/reusable_component/reusables.dart`	1	**Unsafe WebView Configuration**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ```

RULE ID	SEVERITY	FILE LOCATION	LINE	FINDING DETAILS & RISK
				Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.
`SEC-WV-002`	HIGH	`lib/screens/account_tab/dpharma_plus/dpharma_plus_plans.dart`	1	**Unsafe WebView Configuration**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` *Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.*
`SEC-WV-002`	HIGH	`lib/screens/cart/cart_list/my_cart_bloc.dart`	1	**Unsafe WebView Configuration**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` *Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.*
`SEC-WV-001`	HIGH	`lib/screens/cart/payment_webview/payment_webview.dart`	1	**Unsafe WebView Configuration**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` *Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend*

RULE ID	SEVERITY	FILE LOCATION	LINE	FINDING DETAILS & RISK
				databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.
`SEC-WV-002`	HIGH	`lib/screens/cart/payment_webview/payment_webview.dart`	1	**Unsafe WebView Configuration**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` *Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.*
`SEC-AND-003`	HIGH	`android/app/src/main/AndroidManifest.xml`	1	**Android Insecure Configuration**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` *Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.*
`SEC-IOS-001`	HIGH	`ios/Runner/Info.plist`	1	**iOS Insecure Configuration**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` *Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.*

RULE ID	SEVERITY	FILE LOCATION	LINE	FINDING DETAILS & RISK
`SEC-IOS-002`	HIGH	`ios/Runner/Info.plist`	1	**iOS Insecure Configuration**: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` *Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.*
`SEC-DEP-INFO`	LOW	`pubspec.yaml`	1	**Vulnerability Database Check Status**: Using built-in offline Dart/Flutter security rules database. Remote live vulnerability database check is not configured. *Risk: None. This is an informational message stating that live CVE lookup from OSV/GitHub Advisory is not active.*

Security Remediation Action Items

- Remove hardcoded credentials from code. Use environment variables during the build process, or retrieve keys dynamically from a secure vault or backend service at runtime. **Good Example:** ```dart import 'package:flutter_secure_storage/flutter_secure_storage.dart'; final storage = FlutterSecureStorage(); final stripeKey = await storage.read(key: 'stripe_api_key'); ```
- Periodically check dependencies using GitHub security alerts or running "dart pub token" or external tools.

3. High-Priority Performance Flaws: Controller in build()

Instantiating controllers inside build() forces memory re-allocations on every frame render, causing UI stutter, dropped text inputs, and lost scroll states.

COMPONENT / AREA	FILE PATH	LINE	ISSUE DESCRIPTION
Pharmacylistandmap	`lib/screens/NearByPharmacy/pharmacylistandmap.dart`	142	Controller Instantiation Inside Widget build() Method
Chat Details	`lib/screens/chat/details/chat_details.dart`	252	Controller Instantiation Inside Widget build() Method

COMPONENT / AREA	FILE PATH	LINE	ISSUE DESCRIPTION
Tabbar	`lib/screens/tabbar/tab/tabbar.dart`	70	Controller Instantiation Inside Widget build() Method
Searched Products Pharm List	`lib/screens/search/searched_products_pharm_list/searched_products_pharm_list.dart`	43	Controller Instantiation Inside Widget build() Method
Home	`lib/screens/home_tab/home/home.dart`	105	Controller Instantiation Inside Widget build() Method
Featured Brand List	`lib/screens/home_tab/featured_brands/featured_brand_list.dart`	35	Controller Instantiation Inside Widget build() Method
Product List	`lib/screens/home_tab/product_list/product_list.dart`	62	Controller Instantiation Inside Widget build() Method
Address List	`lib/screens/home_tab/address_list/address_list.dart`	94	Controller Instantiation Inside Widget build() Method
Select Map Location	`lib/screens/home_tab/address_list/select_map_location.dart`	79	Controller Instantiation Inside Widget build() Method
Product Other Sellers	`lib/screens/home_tab/product_details/product_other_sellers/product_other_sellers.dart`	41	Controller Instantiation Inside Widget build() Method
Product Review List	`lib/screens/home_tab/product_details/review_list/product_review_list.dart`	49	Controller Instantiation Inside Widget

COMPONENT / AREA	FILE PATH	LINE	ISSUE DESCRIPTION
			build() Method
My Messages List	`lib/screens/account_tab/my_messages/my_messages_list.dart`	58	Controller Instantiation Inside Widget build() Method
Refer Friend	`lib/screens/account_tab/refer_friend/refer_friend.dart`	53	Controller Instantiation Inside Widget build() Method
My Prescriptions	`lib/screens/account_tab/my_prescriptions/list/my_prescriptions.dart`	40	Controller Instantiation Inside Widget build() Method
Manage Address	`lib/screens/account_tab/manage_address/manage_addresses/manage_address.dart`	37	Controller Instantiation Inside Widget build() Method
Order Ticket List	`lib/screens/account_tab/order_related_queries/order_ticket_list.dart`	60	Controller Instantiation Inside Widget build() Method
My Wishlist	`lib/screens/account_tab/my_wishlist/my_wishlist.dart`	39	Controller Instantiation Inside Widget build() Method
Notification List	`lib/screens/account_tab/notifications/notification_list.dart`	43	Controller Instantiation Inside Widget build() Method
My Cart	`lib/screens/cart/cart_list/my_cart.dart`	124	Controller Instantiation Inside Widget build() Method

COMPONENT / AREA	FILE PATH	LINE	ISSUE DESCRIPTION
Pres Order List	`lib/screens/my_orders_tab/pres_order_list/pres_order_list.dart`	92	Controller Instantiation Inside Widget build() Method
My Orders	`lib/screens/my_orders_tab/orders/list/my_orders.dart`	55	Controller Instantiation Inside Widget build() Method
Select Order	`lib/screens/my_orders_tab/create_new_ticket/select_order.dart`	39	Controller Instantiation Inside Widget build() Method
Category List	`lib/screens/category_tab/category_list/category_list.dart`	44	Controller Instantiation Inside Widget build() Method

Standard Fix Architecture:

```
class _SafeInputWidgetState extends State<SafeInputWidget> {
  late final TextEditingController _controller;

  @override
  void initState() {
    super.initState();
    _controller = TextEditingController(); // Allocated once
  }

  @override
  void dispose() {
    _controller.dispose(); // Safely freed from memory
    super.dispose();
  }

  @override
  Widget build(BuildContext context) => TextField(controller: _controller);
}
```

4. Memory & Resource Leaks

ISSUE TYPE	FILE PATH	LINE	HAZARD
`MEM-001 (HIGH)`	`lib/utils/helpers/push_notification_helper.dart`	441	Unclosed StreamSubscription

ISSUE TYPE	FILE PATH	LINE	HAZARD
MEM-002 (HIGH)	`lib/utils/helpers/pref_navigator_observer.dart`	65	Unclosed Timer
MEM-003 (HIGH)	`lib/screens/NearByPharmacy/pharmacylistandmap_bloc.dart`	16	AnimationController Not Disposed
MEM-003 (HIGH)	`lib/screens/order_via_prescription/upload_prescription/upload_prescription.dart`	32	AnimationController Not Disposed
MEM-003 (HIGH)	`lib/screens/order_via_prescription/upload_prescription/upload_prescription_bloc.dart`	19	AnimationController Not Disposed
MEM-003 (HIGH)	`lib/screens/auth/verify_otp/verify_otp_bloc.dart`	22	AnimationController Not Disposed
MEM-003 (HIGH)	`lib/screens/auth/complete_profile/complete_profile.dart`	30	AnimationController Not Disposed
MEM-003 (HIGH)	`lib/screens/auth/complete_profile/complete_profile_bloc.dart`	20	AnimationController Not Disposed
MEM-003 (HIGH)	`lib/screens/auth/login/login_bloc.dart`	22	AnimationController Not Disposed
MEM-003 (HIGH)	`lib/screens/auth/login/login.dart`	29	AnimationController Not Disposed
MEM-003 (HIGH)	`lib/screens/search/search_list/search_list_bloc.dart`	22	AnimationController Not Disposed
MEM-003 (HIGH)	`lib/screens/home_tab/featured_brands/featured_brand_list.dart`	23	AnimationController Not Disposed
MEM-003 (HIGH)	`lib/screens/home_tab/featured_brands/featured_brand_list_bloc.dart`	17	AnimationController Not Disposed
MEM-003 (HIGH)	`lib/screens/home_tab/search_addresses/search_addresses_bloc.dart`	17	AnimationController Not Disposed
MEM-003 (HIGH)	`lib/screens/home_tab/product_list/product_list_bloc.dart`	48	AnimationController Not Disposed
MEM-003 (HIGH)	`lib/screens/account_tab/my_messages/my_messages_list_bloc.dart`	19	AnimationController Not Disposed
MEM-003 (HIGH)	`lib/screens/account_tab/refer_friend/refer_friend_bloc.dart`	15	AnimationController Not Disposed
MEM-003 (HIGH)	`lib/screens/account_tab/edit_profile/edit_profile_bloc.dart`	20	AnimationController Not Disposed
MEM-003 (HIGH)	`lib/screens/account_tab/edit_profile/edit_profile.dart`	35	AnimationController Not Disposed
MEM-003 (HIGH)	`lib/screens/account_tab/manage_address/save_address/save_address_bloc.dart`	39	AnimationController Not Disposed

ISSUE TYPE	FILE PATH	LINE	HAZARD
MEM-003 (HIGH)	lib/screens/account_tab/manage_address/save_address/save_address_bloc.dart	28	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/account_tab/manage_address/manage_addresses/manage_address.dart	26	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/account_tab/order_related_queries/order_ticket_list_bloc.dart	17	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/cart/cart_list/my_cart.dart	51	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/cart/coupons/coupon_list_bloc.dart	15	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/cart/coupons/coupon_list.dart	27	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/my_orders_tab/order_details/order_details_order_placed.dart	1112	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/my_orders_tab/pres_order_list/pres_order_list.dart	35	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/my_orders_tab/rate_product/rate_product_bloc.dart	21	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/my_orders_tab/rate_product/rate_product.dart	30	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/my_orders_tab/orders/list/my_orders_bloc.dart	14	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/my_orders_tab/rate_seller/rate_seller.dart	30	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/my_orders_tab/rate_seller/rate_seller_bloc.dart	19	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/my_orders_tab/create_new_ticket/create_new_ticket_bloc.dart	17	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/my_orders_tab/create_new_ticket/select_order_bloc.dart	14	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/my_orders_tab/create_new_ticket/create_new_ticket.dart	39	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/category_tab/category_list/category_list_bloc.dart	16	AnimationController Not Disposed

5. Class Bloat & Technical Debt (Files > 500 LOC)

FILE LOCATION	LINES	CATEGORY / FOCUS AREA	ARCHITECTURAL ACTION
`lib/utils/common/constants.dart`	1108	Utility Helpers	Split large files into smaller focused units
`lib/utils/helpers/push_notification_helper.dart`	682	Utility Helpers	Split large files into smaller focused units
`lib/reusable_component/toast/flushbar.dart`	793	General	Split large files into smaller focused units
`lib/reusable_component/reusables.dart`	1187	General	Split large files into smaller focused units
`lib/screens/NearByPharmacy/pharmacylistandmap.dart`	639	UI / View	Split large files into smaller focused units
`lib/screens/order_via_prescription/upload_prescription/upload_prescription.dart`	606	UI / View	Split large files into smaller focused units
`lib/screens/chat/details/chat_details.dart`	741	UI / View	Split large files into smaller focused units
`lib/screens/search/search_list/search_list.dart`	722	UI / View	Split large files into smaller focused units
`lib/screens/home_tab/home/home.dart`	1337	UI / View	Split large files into smaller focused units
`lib/screens/home_tab/product_list_filter/product_list_filter.dart`	665	UI / View	Split large files into smaller focused units
`lib/screens/home_tab/product_list_filter/product_list_filter_bloc.dart`	670	UI / View	Split large files into smaller focused units
`lib/screens/home_tab/search_addresses/search_addresses_bloc.dart`	-	UI / View	Save references to all subscriptions created with \`.listen()\` and invoke \`.cancel()\` inside \`dispose()\` for Widgets, or \`onClose()\` for GetxControllers. **Good Example:** \`\`\`dart class _MyState extends State { late StreamSubscription _subscription; @override void initState() { super.initState(); _subscription = myStream.listen((data) { setState(() {}); }); } @override void dispose() { _subscription.cancel(); super.dispose(); } } \`\`\`
`lib/screens/home_tab/product_list/product_list.dart`	732	UI / View	Split large files into smaller focused units

FILE LOCATION	LINES	CATEGORY / FOCUS AREA	ARCHITECTURAL ACTION
`lib/screens/home_tab/address_list/address_list.dart`	830	UI / View	Split large files into smaller focused units
`lib/screens/home_tab/product_details/details/product_details.dart`	1285	UI / View	Split large files into smaller focused units
`lib/screens/account_tab/edit_profile/edit_profile.dart`	591	UI / View	Split large files into smaller focused units
`lib/screens/account_tab/manage_address/save_address/save_address.dart`	852	UI / View	Split large files into smaller focused units
`lib/screens/account_tab/profile_list/profile_list.dart`	523	UI / View	Split large files into smaller focused units
`lib/screens/cart/cart_list/my_cart.dart`	1908	UI / View	Split large files into smaller focused units
`lib/screens/cart/cart_list/my_cart_bloc.dart`	632	UI / View	Split large files into smaller focused units
`lib/screens/my_orders_tab/order_details/order_details_order_placed.dart`	1717	UI / View	Split large files into smaller focused units
`lib/screens/my_orders_tab/pres_order_list/pres_order_list.dart`	911	UI / View	Split large files into smaller focused units
`lib/screens/my_orders_tab/orders/filter/my_orders_filter.dart`	515	UI / View	Split large files into smaller focused units

6. Unresolved Production TODOs

FILE PATH	LINE	MISSING FEATURE / INCOMPLETE CODE
`lib/reusable_component/pdf_view.dart`	25	Declared widget/class/function has no references
`lib/reusable_component/pinch_zoom_page.dart`	26	Declared widget/class/function has no references
`lib/screens/NearByPharmacy/pharmacylistandmap.dart`	51	Declared widget/class/function has no references
`lib/screens/splash/intro_page.dart`	33	Declared widget/class/function has no references
`lib/screens/auth/complete_profile/complete_profile.dart`	38	Declared widget/class/function has no references
`lib/screens/search/searched_products_pharm_list/searched_products_pharm_list.dart`	29	Declared widget/class/function has no references
`lib/screens/search/Pharmacy_products_list_add_cart/Pharmacy_products_list_add_cart.dart`	33	Declared widget/class/function has no references

FILE PATH	LINE	MISSING FEATURE / INCOMPLETE CODE
`lib/screens/home_tab/home/home_bloc.dart`	40	Declared widget/class/function has no references
`lib/screens/home_tab/product_list_filter/product_list_filter_bloc.dart`	441	Declared widget/class/function has no references
`lib/screens/home_tab/featured_brands/featured_brand_list.dart`	28	Declared widget/class/function has no references
`lib/screens/home_tab/search_addresses/search_addresses.dart`	32	Declared widget/class/function has no references
`lib/screens/home_tab/address_list/select_map_location.dart`	46	Declared widget/class/function has no references
`lib/screens/home_tab/product_details/details/product_details.dart`	37	Declared widget/class/function has no references
`lib/screens/home_tab/product_details/product_other_sellers/product_other_sellers.dart`	27	Declared widget/class/function has no references
`lib/screens/home_tab/product_details/review_list/product_review_list.dart`	32	Declared widget/class/function has no references
`lib/screens/account_tab/refer_friend/refer_friend.dart`	30	Declared widget/class/function has no references
`lib/screens/account_tab/manage_address/manage_addresses/manage_address.dart`	30	Declared widget/class/function has no references
`lib/screens/account_tab/profile_list/profile_list.dart`	43	Declared widget/class/function has no references
`lib/screens/my_orders_tab/rate_product/rate_product_bloc.dart`	26	Declared widget/class/function has no references
`lib/screens/my_orders_tab/rate_product/rate_product.dart`	35	Declared widget/class/function has no references
`lib/screens/my_orders_tab/rate_seller/rate_seller.dart`	35	Declared widget/class/function has no references
`lib/screens/my_orders_tab/rate_seller/rate_seller_bloc.dart`	24	Declared widget/class/function has no references

7. Recommended Action Plan

Phase 1 — Hotfix (Days 1–2):

- Remove hardcoded credentials from code. Use environment variables during the build process, or retrieve keys dynamically from a secure vault or backend service at runtime. **Good Example:** ```dart import 'package:flutter_secure_storage/flutter_secure_storage.dart'; final storage = FlutterSecureStorage(); final stripeKey = await storage.read(key: 'stripe_api_key'); ```
- Remove hardcoded credentials from code. Use environment variables during the build process, or retrieve keys dynamically from a secure vault or backend service at runtime. **Good Example:** ```dart import 'package:flutter_secure_storage/flutter_secure_storage.dart'; final storage = FlutterSecureStorage(); final stripeKey = await storage.read(key: 'stripe_api_key'); ```
- Remove hardcoded credentials from code. Use environment variables during the build process, or retrieve keys dynamically from a secure vault or backend service at runtime. **Good Example:** ```dart import 'package:flutter_secure_storage/flutter_secure_storage.dart'; final storage = FlutterSecureStorage(); final stripeKey = await storage.read(key: 'stripe_api_key'); ```

Phase 2 — Performance & Memory (Days 3–5):

- Convert the widget to a `StatefulWidget` and initialize the controller inside `initState()`. Ensure you also call `.dispose()` on it inside the `dispose()` override. **Good Example:** ```dart class MyInputWidget extends StatefulWidget { const MyInputWidget({super.key}); @override State createState() => _MyInputWidgetState(); } class _MyInputWidgetState extends State { late final TextEditingController _textController; @override void initState() { super.initState(); _textController = TextEditingController(); } @override void dispose() { _textController.dispose(); super.dispose(); } @override Widget build(BuildContext context) { return TextField(controller: _textController); } } ```
- Convert the widget to a `StatefulWidget` and initialize the controller inside `initState()`. Ensure you also call `.dispose()` on it inside the `dispose()` override. **Good Example:** ```dart class MyInputWidget extends StatefulWidget { const MyInputWidget({super.key}); @override State createState() => _MyInputWidgetState(); } class _MyInputWidgetState extends State { late final TextEditingController _textController; @override void initState() { super.initState(); _textController = TextEditingController(); } @override void dispose() { _textController.dispose(); super.dispose(); } @override Widget build(BuildContext context) { return TextField(controller: _textController); } } ```
- Save references to all subscriptions created with `.listen()` and invoke `.cancel()` inside `dispose()` for Widgets, or `onClose()` for GetxControllers. **Good Example:** ```dart class _MyState extends State { late StreamSubscription _subscription; @override void initState() { super.initState(); _subscription = myStream.listen((data) { setState(() {}); }); } @override void dispose() { _subscription.cancel(); super.dispose(); } } ```

Phase 3 — Refactoring & Debt (Week 2):

- Split large files into smaller focused units
- Split large files into smaller focused units
- Address pending TODO comments in lib/reusable_component/pdf_view.dart at line 25.

8. Analysis Criteria Observed

The project is audited against the following rules:

Code Quality

- **Proper use of const and final:** If widget not updated once declear and vaeriable value will never change once declear then use const and final with decration (Threshold: const_possible == true assignment_count == 1)
- **Large File LOC:** Trigger when a file has more than 500 non-empty, non-comment code lines (Threshold: file_code_loc > 500)
- **Large Class LOC:** Trigger when a class exceeds configured LOC threshold (Threshold: class_code_loc > 300)
- **Large Function LOC:** Trigger when a function/method exceeds configured LOC threshold (Threshold: function_code_loc > 80)
- **Unused Widgets/Classes/Functions:** Declared widget/class/function has no references (Threshold: reference_count == 0)
- **Unused Imports:** Imported symbol is never referenced (Threshold: import_reference_count == 0)
- **Unreachable Branch:** Branch can never execute based on static control-flow analysis (Threshold: unreachable == true)
- **High Cyclomatic Complexity:** Function exceeds complexity threshold (Threshold: complexity > 10)
- **Duplicate Code:** Similar code block exceeds configured similarity threshold (Threshold: similarity >= 0.85)
- **Large Build Method:** Build method exceeds configured size (Threshold: build_loc > 80)
- **Deep Widget Nesting:** Widget tree nesting exceeds threshold (Threshold: depth > 8)
- **Broad State Rebuild:** State update potentially rebuilds unnecessarily large widget subtree (Threshold: Configurable)
- **Missing Const Opportunity:** Constructor/widget can safely be const but is not (Threshold: const_possible == true)

Security

- **Hardcoded API Key:** API key pattern detected in source/config (Threshold: Pattern + entropy score)
- **Payment Gateway Credential:** Payment provider private/secret credential detected (Threshold: Provider pattern)
- **Firebase Service Account:** Firebase service-account JSON/private key detected (Threshold: Credential pattern)
- **Private Key/Certificate:** PEM/private key/signing material detected (Threshold: private_key_detected == true)
- **Insecure Token Storage:** Access/refresh tokens stored in SharedPreferences/plain files (Threshold: Sensitive token + insecure storage)
- **HTTP/Cleartext Traffic:** Non-TLS HTTP endpoint or cleartext platform configuration detected (Threshold: http:// / cleartext enabled)
- **Certificate Validation Bypass:** Client bypasses certificate validation (Threshold: Trust-all callback/API)
- **Unsafe WebView Configuration:** JavaScript/file access/untrusted navigation enabled without controls (Threshold: Risk pattern)
- **Unsafe Dynamic Data Handling:** Unsafe eval-like/dynamic deserialization or execution pattern detected (Threshold: Risk API pattern)
- **Sensitive Data in Logs:** Token/password/payment/PII passed to logging APIs (Threshold: Sensitive variable/pattern)
- **Android Insecure Configuration:** Debuggable release config, exported components or permissive network settings (Threshold: Risk configuration)
- **iOS Insecure Configuration:** ATS exceptions, insecure entitlements or overly permissive settings (Threshold: Risk configuration)
- **Overly Permissive Firebase Rules:** Firestore/Realtime Database allows broad unauthenticated read/write (Threshold: Broad read/write access)

- **Improper final Usage:** Local variable never reassigned and can be final (Threshold: assignment_count == 1)
- **Unsafe Late Usage:** late variable may be accessed before initialization or is unnecessary (Threshold: Configurable)
- **Naming Convention Violation:** Classes, variables, methods/files violate Dart naming conventions (Threshold: Dart convention)
- **Empty Catch Block:** Catch block has no meaningful handling/logging/rethrow (Threshold: catch_body_effective_statements == 0)

Memory Leak

- **Unclosed StreamSubscription:** StreamSubscription created without matching cancellation/disposal (Threshold: cancel_missing == true)
- **Unclosed Timer:** Timer created without cancellation (Threshold: cancel_missing == true)
- **AnimationController Not Disposed:** AnimationController created in State without dispose (Threshold: dispose_missing == true)
- **TextEditingController Not Disposed:** TextEditingController retained without disposal (Threshold: dispose_missing == true)
- **Scroll/Page Controller Not Disposed:** ScrollController/PageController created without disposal (Threshold: dispose_missing == true)
- **Listener Not Removed:** addListener/addObserver without matching remove (Threshold: remove_missing == true)
- **Unnecessary StatefulWidget:** StatefulWidget has no mutable state or lifecycle requirement (Threshold: stateful_need == false)
- **Missing Const StatelessWidget:** Immutable widget does not use const constructor (Threshold: const_possible == true)
- **Long-Lived Closure Captures State/Context:** Long-lived callback captures State/BuildContext (Threshold: Risk pattern)
- **Unbounded In-Memory Collection:** List/map/cache grows without size/eviction control (Threshold: Configurable size/eviction rule)

Runtime Analysis

- **Heap Growth Over Time:** Heap increases across repeated flows and does not return toward baseline (Threshold: Configurable growth %)
- **Widget/Object Retention:** Objects remain retained after route/widget disposal (Threshold: Retained after disposal)
- **Frame Rendering/Jank:** Frame build/raster time exceeds threshold or dropped frames detected (Threshold: frame_time > 16.67ms)
- **Excessive Widget Rebuilds:** Widget rebuild count significantly exceeds expected baseline (Threshold: Configurable multiplier)
- **Slow Startup:** Cold-start time exceeds configured threshold (Threshold: startup_ms > 3000)
- **Slow API Request:** API request exceeds latency threshold (Threshold: latency_ms > 2000)
- **Excessive Network Traffic:** Unexpected request frequency or payload volume (Threshold: Configurable request/payload threshold)
- **Unhandled Exception:** Unhandled Dart/Flutter exception during test flow (Threshold: unhandled_exception == true)
- **Resource Retained After Screen Exit:** Subscription/timer remains active after navigation away (Threshold: Active resource after dispose)
- **Sensitive Runtime Storage:** Token/payment/PII written to logs or insecure storage (Threshold: Sensitive data detected)

Dependencies

- **Vulnerable Package Version:** Direct/transitive dependency has known security advisory (Threshold: Known CVE/advisory)
- **Unused Dependency:** Declared package has no project references (Threshold: reference_count == 0)
- **Unmaintained Package:** Package has stale releases or weak maintenance signals (Threshold: Configurable inactivity period)

Architecture

- **Layer Violation:** UI directly accesses infrastructure/data layer against selected architecture (Threshold: Forbidden dependency)
- **Mixed State Management:** Conflicting state-management patterns used across modules (Threshold: Configured allowed patterns)
- **Business Logic Inside Widget:** Networking/business rules embedded in build/UI classes (Threshold: Configurable complexity/API count)

Reporting

- **File/Line/Code Evidence:** Every finding must include exact source location and evidence snippet (Threshold: Required fields)
- **Severity/Confidence Scoring:** Every finding receives severity and confidence (Threshold: Required fields)
- **Suppression/Ignore Support:** Approved findings can be suppressed with a reason (Threshold: Suppression requires reason)

Local Threshold Parameters

- **Max Lines of Code:** 500
- **Max Function Lines:** 50
- **Allow TODO:** false
- **Enforce class prefixes:** true